

SUMMARY

Data Scientist and Linux Administrator with 10 years of extensive experience in the IT industry, specializing in deep learning and generative AI for the past 5 years. Proven expertise in designing and deploying AI models, managing cloud infrastructures, and optimising system performance. Skilled in a wide range of technologies and tools including AWS, Azure, Python, and various machine learning frameworks. Adept at leading end-to-end projects from concept to deployment, delivering innovative solutions to complex problems.

KEY SKILLS

Cloud Platforms: AWS (EC2, S3, SageMaker, IAM, Route 53, Amazon Lex, Lambda), Azure (Azure AI Studio, Azure ML Series, Azure Open AI), Oracle Cloud Infrastructure (Oracle Digital Assistant)

Programming and Scripting: Python (Numpy, Pandas, Scikit-learn, Matplotlib, Seaborn), Flask API, Shell Scripting, HTML, CSS

Databases: MySQL, PostgreSQL, MongoDB, Hadoop, Spark

Generative AI: LLM, Gemini-Pro, LLAMA index, OpenAI, GPT-3.5, GPT-4, DALL-E, Text Generation, Document Summarisation, Fine Tuning Models, Vector DB, LangChain, Transformers, BERT, GAN, Encoders and Decoders, Text/Image Generation and Classification, Claude 2, Vector Search, Retrieval Augmented Generation (RAG), Inferencing Models from Hugging Face, Azure Open AI, and GeminiAI

Machine Learning: Statistical Analysis, Data Analysis, Data Modelling, Model Training, Exploratory Data Analysis (EDA), Feature Engineering, Data Visualisation, Regression, Classification, Clustering, Ensemble Techniques, Random Forest, XGBoost

Deep Learning: Keras, PyTorch, TensorFlow, OpenCV, CNN, RNN, LSTM, Transformers, Image Processing, Augmentation Techniques, Computer Vision, Neural Networks, Transfer Learning

MLOps: API Development, Containerisation, Dockerisation, Orchestration, Git, GitHub, Jenkins, Cloud Integrations, Deployment in Cloud and On-premises

Web Frameworks: Flask, Streamlit

Linux Administration: Linux Server Management, Network Configuration, Security and Access Control, System Monitoring, Performance Tuning, Virtualization, VMWare, VirtualBox, Backup and Recovery, Cisco Switching, Firewall Configuration.

TECHNICAL SKILLS

Programming Languages: Python, Shell Scripting

Data Analysis & Visualisation: Pandas, Numpy, Matplotlib, Seaborn

Machine Learning & Deep Learning: Scikit-learn, TensorFlow, Keras, PyTorch, YOLO

Cloud Platforms: AWS, Azure, Oracle Cloud Infrastructure (OCI)

Database Management: MySQL, PostgreSQL, MongoDB

Version Control & CI/CD: Git, GitHub, Jenkins, Docker

KEY DATA SCIENCE PROJECTS

Oracle Digital Assistant (ODA) development for slack channel

- Implemented a NLP based Oracle Digital Assistant solution for the Slack channel to automate some tasks and the responses to the user posts.
- Developed APIs and integrated them with ODA to automate actions and deployed in to Oracle Cloud Infrastructure (OCI)

Generative AI Application for Vehicle Information Extraction

- Developed an application using Gemini AI and prompt engineering techniques to extract vehicle information (license plate number, vehicle colour, type) from images.
- Developed an API for this project and deployed the application on Azure VM, enabling automated data storage into a database.

Generative AI Application for PAN and Aadhaar Information Extraction

- Build an application with Gemini AI to extract PAN and Aadhaar details from images for use in loan applications.
- Developed an API for this project and deployed the solution on Azure VM, automating data extraction and verification processes.

Deep Learning Project for ANPR

- Implemented a deep learning solution using YOLOv8 and transfer learning techniques to extract vehicle information from ANPR camera images.
- Integrated OCR for accurate text extraction from license plates.

Generative AI Application for Itinerary preparation for a trip plan

- Built an application with Gemini AI to prepare an Itinerary for a trip plan to show the best possible Hotel and Restaurants based on user interests and the Hotel offerings.

- Developed an API for this project and deployed the solution on AWS EC2 Linux instance.

PROFESSIONAL EXPERIENCE

Senior Data Scientist

Sep '18 - Present

Oracle IN Pvt Ltd

Hyderabad, IN

Data Science and Machine Learning:

- Led the development and deployment of AI models for various applications, utilising advanced machine learning and deep learning techniques.
- Conducted statistical analysis, data modelling, and model training, focusing on regression, classification, and clustering tasks.
- Implemented ensemble techniques such as Random Forest and XGBoost to improve model accuracy.
- Carried out exploratory data analysis (EDA) and feature engineering to enhance model performance.
- Built and tuned models using frameworks like TensorFlow, Keras, PyTorch, and Scikit-learn.
- Managed model training, monitoring, and tuning, optimising performance on NVIDIA GPUs.

Deep Learning and Generative AI:

- Developed deep learning models using CNNs, RNNs, LSTMs, and Transformers for image processing and text generation.
- Applied augmentation techniques and transfer learning to improve computer vision tasks.
- Created generative AI applications using techniques like GANs, BERT, and various OpenAI models (ChatGPT, Davinci, Curie, Babbage, ADA).
- Executed fine-tuning of models and employed prompt engineering techniques to enhance accuracy and performance.

Data Engineering and MLOps:

- Developed and implemented data processing pipelines for handling structured and unstructured data.
- Ensured efficient data cleaning, preprocessing, and handling of sophisticated datasets.
- Integrated AI/ML solutions into cloud environments, ensuring seamless deployment and orchestration using Docker, Jenkins, and Git.
- Built APIs and managed containerization and orchestration to facilitate scalable deployments in cloud and on-prem environments.

Cloud Infrastructure Management:

- Configured and maintained cloud infrastructures on AWS, Azure, and GCP, utilizing services such as EC2, S3, SageMaker, Azure AI Studio, and Vertex AI.
- Implemented security measures, network configuration, and system monitoring to ensure high availability and performance.

Database and Big Data Tools:

- Managed databases including MySQL, PostgreSQL, and MongoDB, ensuring efficient data storage and retrieval.
- Utilised Big Data tools like Hadoop and Spark for large-scale data processing and analysis.

Web Development and Visualisation:

- Developed web applications and APIs using Flask and Streamlit for data visualization and interactive model deployment.
- Created data visualisations using Matplotlib, Seaborn, and other visualisation tools to present insights effectively.

Linux Administration:

- Managed and maintained Linux-based servers and network infrastructure, ensuring optimal performance, security, and availability.
- Configured and troubleshooted network devices, including routers, switches.
- Implemented automation scripts using Bash and Python, resulting in improved efficiency and reduced manual tasks.
- Monitored system and network performance, identifying and resolving issues to minimize downtime.
- Implemented security measures and access controls to protect sensitive data and network resources.
- Conducted regular system backups and developed disaster recovery strategies.

Senior Network Engineer L3

Oct '17 - Aug '18

Tata Consultancy Services

Hyderabad, IN

- Configuration and maintenance of Cisco routers and switches.
- Serving Cisco clients.
- Practical experience with Cisco network hardware including Cisco 6509, Nexus 7k switches, and Cisco ASR routers 1k and 9k.
- Hands-on experience in network incident management.
- Strong analytical and organizational skills, with the ability to collaborate effectively with multiple geographically dispersed teams.

Senior NOC Executive

Aug '15 - Sep '17

DVL Groups

Hyderabad, IN

- Managed On-premises and cloud infrastructure
- Installation, configure linux, windows servers, managing web servers hosted in AWS and other Data centers
- Site-to-Site VPN configuration.
- Web applications deployments in to cloud.
- development and production environment creation and maintain.
- working with mongoDB and MySQL, NodeJs servers.
- Application and nodes monitoring

Junior GIS Engineer

CYIENT Ltd

Dec '12 - May '15

Hyderabad, IN

- Processed and classified traffic signs from input data captured by geospatial vehicles on the road. Integrated classified data into GIS applications to provide driver guidance and enhance navigation systems.
- Ensuring the accuracy and quality of spatial data and GIS applications.
- Developing custom GIS tools and scripts to automate tasks or enhance GIS capabilities.
- Creating maps and visualisations based on spatial data.

EDUCATION

Pursuing Master's in Data Science

Liverpool John Moores University

Post Graduation in Data Science

IIIT Bangalore

Bachelor science in Computers and Statistics

Andhra University

Oct '23 - Sep '25

Sep '22 - Oct '23

Mar '08 - Mar '12

Vizianagaram, IN

CERTIFICATIONS

- CCNA and CCNP (Routing and Switching) (CSCO12804886)

ADDITIONAL INFORMATION

- **Languages:** English (Professional), Telugu (Native), Hindi (Beginner)
- **Interests:** Learning new technologies